

# Daily Problem Solutions Booklet

*Comprehensive Step-by-Step Analytical Solutions*

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## Integrals

### 1.1 Medium Integral

#### Lambert W-Function Integral

Evaluate the definite integral:

$$\int_{-1/e}^{\infty} \frac{W(e^{-W(x)})^4}{W(x) + 1} dx$$

**Solution:** Let  $u = W(x)$ , which by definition means  $x = ue^u$ . Differentiating both sides gives  $dx = (u + 1)e^u du$ . For the limits of integration, as  $x \rightarrow -1/e$ ,  $u \rightarrow -1$ , and as  $x \rightarrow \infty$ ,  $u \rightarrow \infty$ . Substituting these into the integral:

$$\int_{-1}^{\infty} \frac{W(e^{-u})^4}{u + 1} (u + 1)e^u du = \int_{-1}^{\infty} W(e^{-u})^4 e^u du$$

Now apply a second substitution. Let  $v = W(e^{-u})$ , which means  $e^{-u} = ve^v$ . Taking the natural logarithm of both sides yields  $-u = \ln(v) + v$ . Differentiating gives  $-du = \left(\frac{1}{v} + 1\right) dv = \frac{v+1}{v} dv$ , or  $du = -\frac{v+1}{v} dv$ . Also note that  $e^u = \frac{1}{ve^v}$ . For the new limits: as  $u \rightarrow -1$ ,  $v = W(e) = 1$ . As  $u \rightarrow \infty$ ,  $v = W(0) = 0$ . Applying this substitution transforms the integral to:

$$\int_1^0 v^4 \left(\frac{1}{ve^v}\right) \left(-\frac{v+1}{v}\right) dv = \int_0^1 \frac{v^3(v+1)}{e^v} dv = \int_0^1 (v^4 + v^3)e^{-v} dv$$

We can compute this using repeated integration by parts or the general formula  $\int x^n e^{-x} dx = -e^{-x} \sum_{k=0}^n \frac{n!}{k!} x^k$ :

$$\int_0^1 v^3 e^{-v} dv = \left[-e^{-v}(v^3 + 3v^2 + 6v + 6)\right]_0^1 = 6 - \frac{16}{e}$$

$$\int_0^1 v^4 e^{-v} dv = \left[-e^{-v}(v^4 + 4v^3 + 12v^2 + 24v + 24)\right]_0^1 = 24 - \frac{65}{e}$$

Adding these results together gives the final exact value:

$$\left(24 - \frac{65}{e}\right) + \left(6 - \frac{16}{e}\right) = 30 - \frac{81}{e}$$

### 1.2 Hard Integral

### Trigonometric Rational Integral

Evaluate the definite integral:

$$\int_0^{\pi} \frac{x^2 \cos(x)(2 \sin^2(x) + \cos^2(x) + 1)}{(\cos^2(x) + 1)^2} dx$$

**Solution:** Let  $I$  be the integral. We first notice that the trigonometric part of the integrand can be simplified. Using  $\sin^2(x) = 1 - \cos^2(x)$ , we have  $2 \sin^2(x) + \cos^2(x) + 1 = 2(1 - \cos^2(x)) + \cos^2(x) + 1 = 3 - \cos^2(x)$ . Now, notice that the derivative of  $h(x) = \frac{\sin(x)}{1 + \cos^2(x)}$  is exactly the trigonometric fraction in the integral:

$$\begin{aligned} h'(x) &= \frac{\cos(x)(1 + \cos^2(x)) - \sin(x)(-2 \cos(x) \sin(x))}{(1 + \cos^2(x))^2} = \frac{\cos(x)(1 + \cos^2(x) + 2 \sin^2(x))}{(1 + \cos^2(x))^2} \\ &= \frac{\cos(x)(3 - \cos^2(x))}{(1 + \cos^2(x))^2} \end{aligned}$$

Therefore, our integral takes the form:

$$I = \int_0^{\pi} x^2 \frac{d}{dx} \left( \frac{\sin(x)}{1 + \cos^2(x)} \right) dx$$

Applying integration by parts with  $u = x^2$  and  $dv = h'(x)dx$ :

$$I = \left[ x^2 \frac{\sin(x)}{1 + \cos^2(x)} \right]_0^{\pi} - \int_0^{\pi} 2x \frac{\sin(x)}{1 + \cos^2(x)} dx$$

Since  $\sin(\pi) = \sin(0) = 0$ , the boundary term evaluates to zero. We are left with:

$$I = -2 \int_0^{\pi} x \frac{\sin(x)}{1 + \cos^2(x)} dx$$

Using the well-known identity  $\int_0^{\pi} x f(\sin(x), \cos^2(x)) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin(x), \cos^2(x)) dx$ , we can pull out a factor of  $\pi/2$ :

$$I = -2 \left( \frac{\pi}{2} \right) \int_0^{\pi} \frac{\sin(x)}{1 + \cos^2(x)} dx = -\pi \int_0^{\pi} \frac{\sin(x)}{1 + \cos^2(x)} dx$$

Now, let  $w = \cos(x) \implies dw = -\sin(x)dx$ . The bounds change from  $[0, \pi]$  to  $[1, -1]$ :

$$I = -\pi \int_1^{-1} \frac{-dw}{1 + w^2} = -\pi \int_{-1}^1 \frac{dw}{1 + w^2}$$

$$I = -\pi \left[ \tan^{-1}(w) \right]_{-1}^1 = -\pi \left( \frac{\pi}{4} - \left( -\frac{\pi}{4} \right) \right) = -\pi \left( \frac{\pi}{2} \right) = -\frac{\pi^2}{2}$$

## Differentials

### 2.1 Medium Differential

#### Locus AI Neural Weight Balancer

The Locus AI neural weight balancer maps a continuous loss boundary  $x$  against an internal epoch scale  $y$ . The algorithm strictly enforces that the marginal rate of change of the epoch scale with respect to the loss boundary  $\frac{dy}{dx}$  is continuously equal to the reciprocal of the exact difference between  $e^{-y}$  and  $x$ . If the loss boundary initializes at exactly  $x = 3$  during epoch  $y = 0$ , what is the exact numerical value of  $x \cdot e^y$  evaluated at epoch  $y = 2$ ?

**Solution:** The problem translates to the following first-order differential equation:

$$\frac{dy}{dx} = \frac{1}{e^{-y} - x}$$

Taking the reciprocal of both sides to view  $x$  as a function of  $y$ :

$$\frac{dx}{dy} = e^{-y} - x \implies \frac{dx}{dy} + x = e^{-y}$$

This is a standard first-order linear differential equation in standard form. The integrating factor is:

$$I(y) = e^{\int 1 dy} = e^y$$

Multiplying the entire equation by the integrating factor gives:

$$e^y \frac{dx}{dy} + e^y x = 1 \implies \frac{d}{dy}(xe^y) = 1$$

Integrating both sides with respect to  $y$  yields the general solution:

$$xe^y = y + C$$

We apply the initial condition: when  $y = 0$ ,  $x = 3$ .

$$(3)e^0 = 0 + C \implies C = 3$$

Therefore, the particular solution governing the system is:

$$xe^y = y + 3$$

We are asked to evaluate the exact numerical value of the expression  $x \cdot e^y$  at epoch  $y = 2$ . Plugging in  $y = 2$ :

$$xe^2 = 2 + 3 = 5$$

### 2.2 Hard Differential

### Complex Bernoulli Differential Equation

Let  $f(x)$  satisfy the differential equation:

$$\frac{dy}{dx} + \left( \frac{2x}{1+x^2} + \frac{3}{x \ln(x)} \right) y = -\frac{1}{4}(1+x^2)^4(\ln(x))^{12}y^5$$

Given the initial condition  $f(e) = 1$ , find the exact value of  $f(10)$ .

**Solution:** This is a Bernoulli differential equation of the form  $\frac{dy}{dx} + P(x)y = Q(x)y^n$  with  $n = 5$ . We apply the standard substitution  $v = y^{1-n} = y^{-4}$ . Differentiating with respect to  $x$  yields  $\frac{dv}{dx} = -4y^{-5}\frac{dy}{dx}$ . First, divide the original equation by  $y^5$ :

$$y^{-5}\frac{dy}{dx} + \left( \frac{2x}{1+x^2} + \frac{3}{x \ln(x)} \right) y^{-4} = -\frac{1}{4}(1+x^2)^4(\ln(x))^{12}$$

Substituting our expressions for  $v$  and its derivative, the equation becomes a linear first-order differential equation:

$$-\frac{1}{4}\frac{dv}{dx} + \left( \frac{2x}{1+x^2} + \frac{3}{x \ln(x)} \right) v = -\frac{1}{4}(1+x^2)^4(\ln(x))^{12}$$

Multiplying by  $-4$ :

$$\frac{dv}{dx} - 4 \left( \frac{2x}{1+x^2} + \frac{3}{x \ln(x)} \right) v = (1+x^2)^4(\ln(x))^{12}$$

The integrating factor  $I(x)$  is computed as:

$$I(x) = \exp \left( \int -4 \left( \frac{2x}{1+x^2} + \frac{3}{x \ln(x)} \right) dx \right) = \exp \left( -4 \ln(1+x^2) - 12 \ln(\ln(x)) \right)$$

$$I(x) = \frac{1}{(1+x^2)^4(\ln(x))^{12}}$$

Multiplying the linear equation by  $I(x)$ :

$$\frac{d}{dx} \left( v \cdot \frac{1}{(1+x^2)^4(\ln(x))^{12}} \right) = 1$$

Integrating with respect to  $x$ :

$$v \cdot \frac{1}{(1+x^2)^4(\ln(x))^{12}} = x + C \implies v(x) = (x+C)(1+x^2)^4(\ln(x))^{12}$$

Since  $v = y^{-4}$ , we have  $y(x) = [(x+C)(1+x^2)^4(\ln(x))^{12}]^{-1/4}$ . Applying the initial condition  $f(e) = 1$ :

$$1 = \left[ (e+C)(1+e^2)^4(\ln(e))^{12} \right]^{-1/4} \implies 1 = (e+C)(1+e^2)^4$$

Solving for C:

$$C = \frac{1}{(1 + e^2)^4} - e = (1 + e^2)^{-4} - e$$

To find  $f(10)$ , substitute  $x = 10$  and C back into the solution for  $y(x)$ :

$$f(10) = \left[ \left( 10 - e + (1 + e^2)^{-4} \right) (101)^4 (\ln(10))^{12} \right]^{-1/4}$$

## Derivatives

### 3.1 Medium Derivative

#### Derivative of a Functional Power

Let  $f(x) = (x^2 + 2)^{\frac{1}{x}}$ . Find  $f'(1)$ .

**Solution:** To find the derivative of a function where both the base and exponent depend on  $x$ , we employ logarithmic differentiation. Taking the natural logarithm of both sides:

$$\ln(f(x)) = \frac{1}{x} \ln(x^2 + 2)$$

Differentiate both sides with respect to  $x$  using the product and chain rules:

$$\frac{f'(x)}{f(x)} = -\frac{1}{x^2} \ln(x^2 + 2) + \frac{1}{x} \left( \frac{2x}{x^2 + 2} \right)$$

Simplify the right hand side:

$$\frac{f'(x)}{f(x)} = -\frac{\ln(x^2 + 2)}{x^2} + \frac{2}{x^2 + 2}$$

To find  $f'(1)$ , we first need the value of  $f(1)$ :

$$f(1) = (1^2 + 2)^{\frac{1}{1}} = 3^1 = 3$$

Now evaluate the derivative expression at  $x = 1$ :

$$\frac{f'(1)}{3} = -\frac{\ln(1^2 + 2)}{1^2} + \frac{2}{1^2 + 2} = -\ln(3) + \frac{2}{3}$$

Multiply by 3 to isolate  $f'(1)$ :

$$f'(1) = 3 \left( \frac{2}{3} - \ln(3) \right) = 2 - 3\ln(3)$$

### 3.2 Hard Derivative

#### 10th Derivative of a Product at Zero

Let  $f(x) = xe^{x^2} \sin(2x)$ . Find  $f^{(10)}(0)$ .

**Solution:** Instead of computing the derivative 10 times, we utilize Maclaurin series expansions. Recall the relation between the  $n$ -th derivative at  $x = 0$  and the coefficient of  $x^n$  in the Maclaurin series:  $c_n = \frac{f^{(n)}(0)}{n!}$ , so  $f^{(10)}(0) = 10! \cdot c_{10}$ . The standard series for the individual components are:

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} + \dots = 1 + x^2 + \frac{1}{2}x^4 + \frac{1}{6}x^6 + \frac{1}{24}x^8 + \dots$$

$$\sin(2x) = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!} + \frac{(2x)^9}{9!} - \dots = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{315}x^7 + \frac{4}{2835}x^9 - \dots$$

We are looking for the  $x^{10}$  term in  $f(x) = xe^{x^2}\sin(2x)$ . This is equivalent to finding the  $x^9$  term in the product  $g(x) = e^{x^2}\sin(2x)$ , since multiplying by the leading  $x$  shifts the degree up by 1. We gather all combinations of terms from  $e^{x^2}$  and  $\sin(2x)$  whose exponents sum to 9: 1.  $x^8$  from  $e^{x^2}$  and  $x^1$  from  $\sin(2x)$ :  $\left(\frac{1}{24}\right) \times (2) = \frac{1}{12}$  2.  $x^6$  from  $e^{x^2}$  and  $x^3$  from  $\sin(2x)$ :  $\left(\frac{1}{6}\right) \times \left(-\frac{4}{3}\right) = -\frac{2}{9}$  3.  $x^4$  from  $e^{x^2}$  and  $x^5$  from  $\sin(2x)$ :  $\left(\frac{1}{2}\right) \times \left(\frac{4}{15}\right) = \frac{2}{15}$  4.  $x^2$  from  $e^{x^2}$  and  $x^7$  from  $\sin(2x)$ :  $(1) \times \left(-\frac{8}{315}\right) = -\frac{8}{315}$  5.  $x^0$  from  $e^{x^2}$  and  $x^9$  from  $\sin(2x)$ :  $(1) \times \left(\frac{4}{2835}\right) = \frac{4}{2835}$  Summing these coefficients gives  $c_{10}$ :

$$c_{10} = \frac{1}{12} - \frac{2}{9} + \frac{2}{15} - \frac{8}{315} + \frac{4}{2835}$$

To combine these, use a common denominator of 11340:

$$c_{10} = \frac{945 - 2520 + 1512 - 288 + 16}{11340} = -\frac{335}{11340} = -\frac{67}{2268}$$

Therefore, the 10th derivative is:

$$f^{(10)}(0) = 10! \left(-\frac{67}{2268}\right) = 3,628,800 \left(-\frac{67}{2268}\right) = 1600 \times (-67) = -107,200$$



## Physics & Engineering

### 4.1 Mechanics

#### Tunnel Through the Earth

A tunnel is dug along a chord of the earth at a perpendicular distance  $R/2$  from the earth's centre. The wall of the tunnel is frictionless. Find the force (in N) exerted by the wall on a particle of mass  $m$  when it is at a distance  $x$  from the centre of the tunnel. Assume  $m = 4.0$  kg,  $x = 1500$  km, the earth's radius  $R = 6400$  km, and the acceleration due to gravity at the earth's surface  $g = 10 \text{ m s}^{-2}$ .

**Solution:** The force exerted by the wall of the tunnel is the normal force, which balances the component of the gravitational force perpendicular to the tunnel. The gravitational field inside a uniform sphere at a distance  $r$  from its center is proportional to  $r$ :

$$g(r) = g \frac{r}{R}$$

The gravitational force on the particle is directed towards the center of the Earth:

$$F_g = mg \frac{r}{R}$$

The distance  $r$  of the particle from the Earth's center is given by the Pythagorean theorem,  $r = \sqrt{x^2 + (R/2)^2}$ . The angle  $\theta$  between the position vector and the perpendicular to the tunnel satisfies  $\cos(\theta) = \frac{R/2}{r}$ . The component of the gravitational force perpendicular to the tunnel must be entirely counteracted by the normal force  $F_N$  from the wall:

$$F_N = F_g \cos(\theta) = \left( mg \frac{r}{R} \right) \left( \frac{R/2}{r} \right) = \frac{1}{2} mg$$

Notably, the normal force is constant and completely independent of the distance  $x$ . Plugging in the given values ( $m = 4.0$  kg and  $g = 10 \text{ m s}^{-2}$ ):

$$F_N = \frac{1}{2} (4.0)(10) = 20 \text{ N}$$

### 4.2 Quantum Mechanics

#### Compton Scattering Wavelength Shift

A high-energy photon of initial wavelength  $\lambda = 500$  pm undergoes Compton scattering from a stationary free electron at an angle of  $\theta = 60^\circ$ . Calculate the final wavelength of the scattered photon in picometers (pm). Take the Compton wavelength of an electron as  $\frac{h}{m_e c} = 2.43$  pm and  $\cos(60^\circ) = 0.5$ .

**Solution:** The shift in wavelength of a photon undergoing Compton scattering is given by the Compton formula:

$$\Delta\lambda = \lambda_f - \lambda_i = \lambda_C(1 - \cos(\theta))$$

Where  $\lambda_C = \frac{h}{m_e c}$  is the Compton wavelength of the electron. We are given  $\lambda_C = 2.43 \text{ pm}$  and  $\cos(60^\circ) = 0.5$ . Substituting these values into the equation:

$$\Delta\lambda = 2.43 \times (1 - 0.5) = 2.43 \times 0.5 = 1.215 \text{ pm}$$

The final wavelength  $\lambda_f$  is the initial wavelength plus the shift:

$$\lambda_f = \lambda_i + \Delta\lambda = 500 \text{ pm} + 1.215 \text{ pm} = 501.215 \text{ pm}$$

### 4.3 Electric Fields

#### Electrostatic Energy of a Dielectric Layer

A point charge  $q = 3.0 \text{ } \mu\text{C}$  is located at the centre of a spherical layer of uniform isotropic dielectric with relative permittivity  $\epsilon = 3.0$ . The inside radius of the layer is equal to  $a = 200 \text{ mm}$ , the outside radius is  $b = 500 \text{ mm}$ . Find the electrostatic energy inside the dielectric layer in mJ. Assume  $\epsilon_0 = \frac{1}{36\pi \times 10^9} \text{ F m}^{-1}$ .

**Solution:** The electric displacement field magnitude  $D$  at a distance  $r$  from the point charge inside the dielectric is given by Gauss's Law:

$$D = \frac{q}{4\pi r^2}$$

The electric field is  $E = \frac{D}{\epsilon_0 \epsilon} = \frac{q}{4\pi \epsilon_0 \epsilon r^2}$ . The electrostatic energy density  $u$  within the dielectric layer is:

$$u = \frac{1}{2} \epsilon_0 \epsilon E^2 = \frac{1}{2} \epsilon_0 \epsilon \left( \frac{q}{4\pi \epsilon_0 \epsilon r^2} \right)^2 = \frac{q^2}{32\pi^2 \epsilon_0 \epsilon r^4}$$

To find the total energy  $W$  contained within the spherical shell, we integrate the energy density over the volume from  $r = a$  to  $r = b$ . The differential volume element is  $dV = 4\pi r^2 dr$ :

$$W = \int_a^b u dV = \int_a^b \frac{q^2}{32\pi^2 \epsilon_0 \epsilon r^4} (4\pi r^2) dr = \int_a^b \frac{q^2}{8\pi \epsilon_0 \epsilon r^2} dr$$

Evaluating the integral:

$$W = \frac{q^2}{8\pi \epsilon_0 \epsilon} \left[ -\frac{1}{r} \right]_a^b = \frac{q^2}{8\pi \epsilon_0 \epsilon} \left( \frac{1}{a} - \frac{1}{b} \right)$$

Let's substitute the given values:  $q = 3.0 \times 10^{-6} \text{ C}$ ,  $\epsilon = 3.0$ ,  $a = 0.2 \text{ m}$ ,  $b = 0.5 \text{ m}$ , and  $\epsilon_0 = \frac{1}{36\pi \times 10^9} \text{ F m}^{-1}$ . Note that  $\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ m F}^{-1}$ , so  $\frac{1}{8\pi \epsilon_0} = 4.5 \times 10^9$ . The geometric factor is  $\frac{1}{0.2} - \frac{1}{0.5} = 5 - 2 = 3 \text{ m}^{-1}$ .

$$W = (3.0 \times 10^{-6})^2 \times (4.5 \times 10^9) \times \left( \frac{1}{3.0} \right) \times 3$$

$$W = (9.0 \times 10^{-12}) \times (4.5 \times 10^9) \times 1 = 40.5 \times 10^{-3} \text{ J}$$

Converting to millijoules gives exactly 40.5 mJ.